

Fueling a Resource Secure Future >>



GMDC's Vision for Energy Security and Viksit Bharat

India stands at the cusp of an industrial and economic transformation, with the vision of Viksit Bharat driving its roadmap for self-reliance, sustainability, and long-term growth. At the core of this ambition lies energy security and access to critical minerals—two pillars that will shape the country's future in power, infrastructure, and emerging technologies.

As one of India's leading PSUs in mining with a strong, proven legacy in lignite, GMDC is playing a pivotal role in strengthening these foundations, ensuring that the nation has the resources it needs to fuel progress.

With the aim to continue contributing to the nation's vision of progress, and with a steadfast commitment to strengthening India's energy future, GMDC has strategically expanded into coal mining in Odisha. GMDC's acquisition of three coal mines, with a combined geological reserve exceeding 2000 million tonnes, ensures a reliable domestic supply & enhanced energy security.

While coal remains critical, the demand for cleaner, low-carbon energy solutions is rapidly growing. GMDC's exploration of Blue Hydrogen marks a pivotal step in India's energy transition. Produced through coal gasification, it offers a scalable and cost-effective fuel alternative for heavy industries, power generation, and mobility, bridging the gap between conventional energy and a cleaner future. Leveraging Odisha's rich coal reserves, GMDC is taking pioneering efforts for Underground

Coal Gasification (UCG)—a process that unlocks deep coal reserves while significantly reducing environmental impact. By integrating hydrogen-based solutions into India's broader energy framework, GMDC is not only opening new avenues for energy, it is also leading India towards a cleaner & greener future.

Considering India's acceleration toward a tech-driven economy, GMDC's plan to mine copper indigenously marks a decisive step in its mission to secure critical resources for the nation's future. Recognizing copper as the backbone of electrical infrastructure, renewable energy, and advanced manufacturing, GMDC is developing the Ambaji Copper Mine in Gujarat's mineral-rich belt. With deposits estimated at USD 3 billion, this project is set to become one of India's most significant copper reserves. GMDC's Copper Project is part of a larger vision, one that aligns with India's ambition to build a self-sufficient, future-ready industrial ecosystem that thrives on secure, sustainable, and strategically developed resources.

Equally important is GMDC's role in Rare Earth Elements (REEs), a category of minerals essential for batteries, semiconductors, and high-performance electronics. The Ambadungar REE deposit in Gujarat is one of India's most promising sources of these critical elements, and GMDC is actively making efforts to develop this resource, positioning itself as a key contributor to India's technological and industrial growth. Given that REEs are currently dominated by a handful of global

suppliers, domestic exploration is vital to ensuring India's competitiveness in high-tech sectors.

As GMDC expands its reach across coal, clean energy, critical minerals, & developing a comprehensive Rare Earths Value Chain, its role extends beyond resource extraction—it is actively shaping India's industrial and economic landscape. These developments are creating employment opportunities, strengthening supply chains, and driving economic growth at both regional and national levels. Through structured CSR initiatives, GMDC is also working toward education, healthcare, and environmental conservation, ensuring that the benefits of industrial progress reach the communities where it operates.

GMDC's evolution is not just about resource expansion—it is about shaping India's future. Every initiative—whether strengthening domestic copper supply, advancing clean energy solutions, or ensuring long-term resource security—is aligned with the nation's goal of Viksit Bharat. The future demands self-reliance, sustainability, and innovation, and GMDC stands at the forefront, driving a future where India is not just resource-rich but resource-secure, resilient, and ready to be the catalyst in India's progress.



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